

GenBI Core Database Schema (SQL Server)

Official Architecture & Reference Guide Adapted for Transact-SQL (T-SQL)

Database Engine	Microsoft SQL Server	Target Database	genbi_db
Primary String Dialect	Unicode (NVARCHAR)	Total Tables	9 Tables (with constraints & seed logic)

1. Architectural Migrations & T-SQL Optimizations

This updated schema has been engineered to comply fully with Transact-SQL (T-SQL) syntax and constraints. Unlike MySQL, Microsoft SQL Server handles unstructured text documents via NVARCHAR(MAX) and enforces strict rules regarding multiple cascading foreign key paths to avoid cyclic dependencies (SQL Server Error 1785).

Cascade Isolation Rule: To satisfy SQL Server's referential integrity architecture, the junction tables query_tables and session_queries utilize ON DELETE NO ACTION on secondary reference tracks. This mitigates multiple activation paths flowing from the core auth identity tracking vector.

2. Table-by-Table Data Dictionary

2.1 Table: auth

Stores user credentials and alphanumeric system identification tokens. Includes pre-seeded test accounts.

Column Name	Data Type	Attributes	Description / Rationale
user_id	VARCHAR(50)	PK NOT NULL	Primary tracking alphanumeric ID (e.g., 'IS1026', 'IS1033').
name	VARCHAR(100)	NOT NULL	User's legal profile display name.
password	VARCHAR(255)	NOT NULL	Secure hash string storage capacity.

2.2 Table: table_info

Maintains structured configurations and semantic rule patterns regarding physical datasources.

Column Name	Data Type	Attributes	Description / Rationale
id	INT	PK IDENTITY(1,1)	Surrogate sequential identifier.
name	VARCHAR(100)	NOT NULL	Logical name of physical table.
user_id	VARCHAR(50)	FK NULL	Owner mapping reference. Cascades on user deletion.
knowledgebase	NVARCHAR(MAX)	NULL	LLM injection context descriptors.
metadata	NVARCHAR(MAX)	NULL	Serialized JSON schema map (validate via ISJSON()).
createdAt	DATETIME	NULL	Chronological profile creation timestamp.

2.3 Table: queries

Historical tracking payload log specifying prompt metrics, compiled SQL, and state change audit strings.

Column Name	Data Type	Attributes	Description / Rationale
id	INT	PK IDENTITY(1,1)	System wide unique query log key.
prompt	NVARCHAR(MAX)	NOT NULL	Raw Natural Language AI command payload.
sql_query	NVARCHAR(MAX)	NOT NULL	LLM-compiled target environment query text.
summary	NVARCHAR(MAX)	NOT NULL	Detailed breakdown summarizing prompt transformation state changes.
updated_columns	NVARCHAR(MAX)	NOT NULL	Explicit logging tracking target columns changed post execution.

2.4 Table: query_tables

Junction structure establishing database table mapping histories against compilation queries.

Column Name	Data Type	Attributes	Description / Rationale
query_id	INT	PK FK	References queries.id (Cascades).
table_id	INT	PK FK	References table_info.id (Isolated: NO ACTION).

2.5 Table: sessions

Captures specific sandbox interactive environment types using strict check constraints.

Column Name	Data Type	Attributes	Description / Rationale
session_id	INT	PK IDENTITY(1,1)	Workspace boundary unique key.
user_id	VARCHAR(50)	FK NULL	References session operator profile.
type	VARCHAR(20)	NOT NULL CHECK	Enforced limits: 'Dashboard' or 'Transformation' .
date_created	DATETIME2	DEFAULT NOW	High precision system deployment creation timestamp.

2.6 Table: session_queries

Audited junction tracing continuous iterations inside execution sessions.

Column Name	Data Type	Attributes	Description / Rationale
session_id	INT	PK FK	References sessions.session_id (Cascades).
query_id	INT	PK FK	References queries.id (Isolated: NO ACTION).
date_created	DATETIME2	DEFAULT NOW	Session execution log entry time tracking.
date_modified	DATETIME2	DEFAULT NOW	Modified dynamically via application layer updates.

2.7 Table: charts

Decoupled data presentation definitions generated by frontend analytical parsing.

Column Name	Data Type	Attributes	Description / Rationale
chart_id	INT	PK IDENTITY(1,1)	Unique surrogate identifier for visualization block.
chart_type	VARCHAR(50)	NOT NULL	Rendering strategy selector string configuration.
prompt	NVARCHAR(MAX)	NOT NULL	Formatting instruction payload targeted at styling properties.
sql_query	NVARCHAR(MAX)	NOT NULL	Data generation routine query text.
metadata	NVARCHAR(MAX)	NULL	JSON parameters dictionary specifying styles/axes variables.

2.8 Table: dashboards

Parent visualization viewport container monitoring state and engine calculation refresh loops.

Column Name	Data Type	Attributes	Description / Rationale
dashboard_id	INT	PK IDENTITY(1,1)	Unique identity primary pointer.
user_id	VARCHAR(50)	FK NULL	References auth.user_id (Cascades).
status	VARCHAR(10)	NOT NULL CHECK	Enforced status constraints: 'Live' or 'Inactive'.
breath_time	INT	NOT NULL DEF 5	Integer frequency parsing counter mapped in whole minutes.

2.9 Table: dashboard_charts

Junction layout pinning multiple decoupled charts inside complex dashboard arrangements.

Column Name	Data Type	Attributes	Description / Rationale
dashboard_id	INT	PK FK	References dashboards.dashboard_id (Cascades).
chart_id	INT	PK FK	References charts.chart_id (Cascades).

3. Key Production Rules for SQL Server (T-SQL)

3.1 JSON Querying via OpenJSON

Since structures utilize `NVARCHAR(MAX)` for handling semi-structured elements (`metadata` and `knowledgebase`), use SQL Server's native json processing commands. You can safely query elements via `JSON_VALUE(metadata, '$.theme')` or validate formats at the persistence boundary using `ISJSON(metadata) = 1`.

3.2 Multiple Cascading Path Exclusions

SQL Server blocks execution if an operation path causes an item deletion to update structural trees along more than one concurrent relational downstream. Because removing a user record deletes both `table_info` and `queries` records, cross-links inside mapping environments are secured using `ON DELETE NO ACTION` to maintain processing safety.